

Day 11 Of Communication System Class

5.1 Multiplexing fundamentals: FDM, TDM, WDM and applications

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As data > channel . so , its important to utilize bandwidth for communication of multi to multi mode .

Techniques Used in channel bandwidth utilization are as :

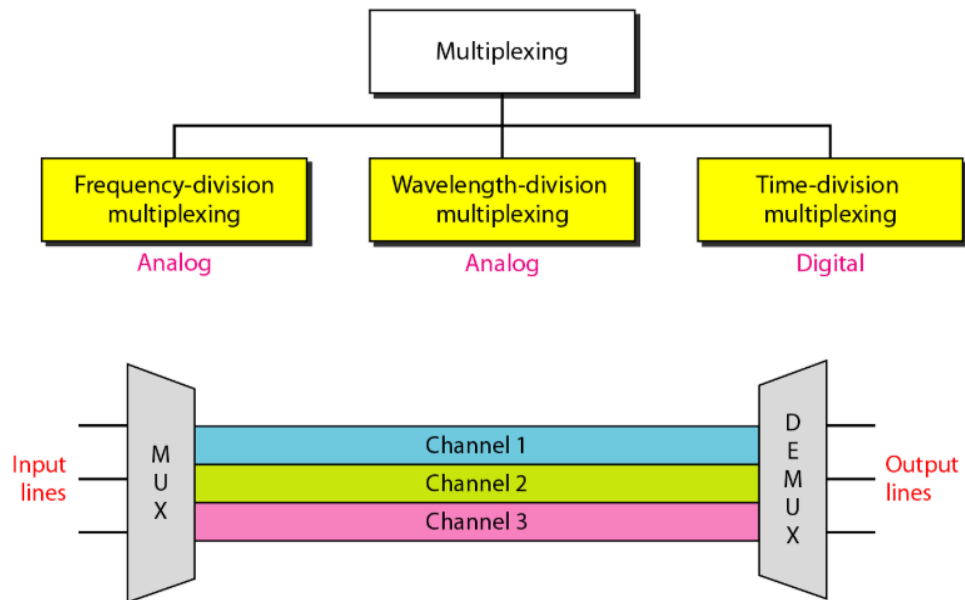
1. Multiplexing
 - 1.1 TDM
 - 1.2 FDM
 - 1.3 WDM
2. Switching
 - 2.1 Circuit Switching
 - 2.2 Message Switching
 - 2.3 Packet Switching
3. Multiple Access
 - 3.1 TDMA
 - 3.2 FDMA
 - 3.3 SDMA
 - 3.4 CDMA

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SOURCE

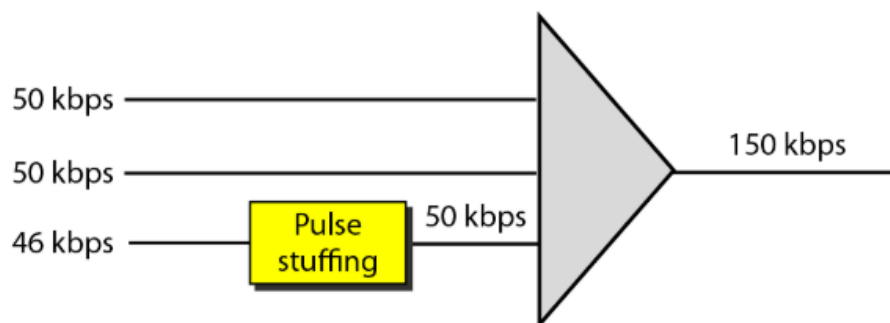
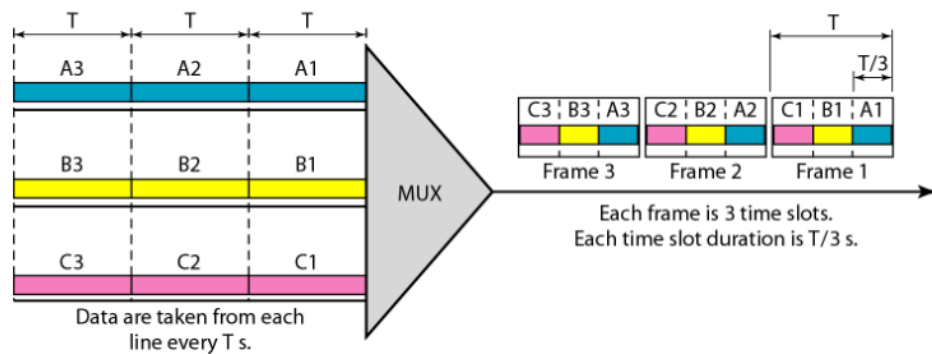
<https://medium.com/@saurabgyawali/how-millions-of-phone-calls-travel-through-a-single-cable-0c792488e393>

1. Multiplexing



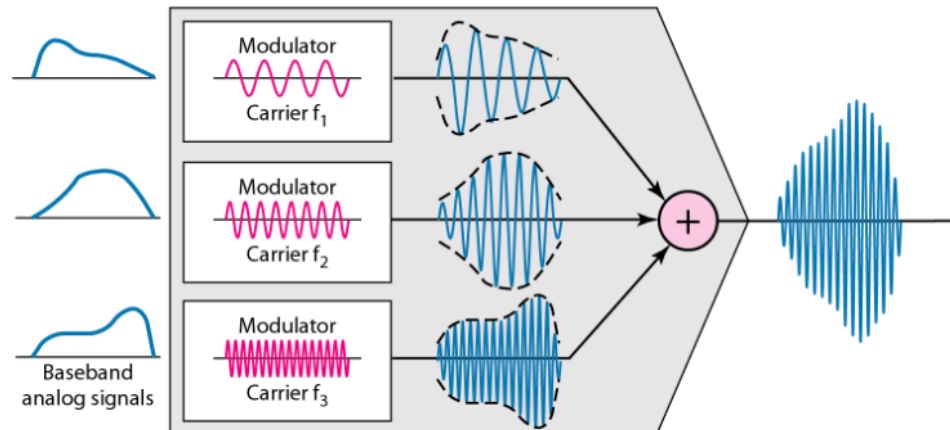
1.1 TDM

TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.



1.2 FDM

FDM is an analog multiplexing technique that combines analog signals.

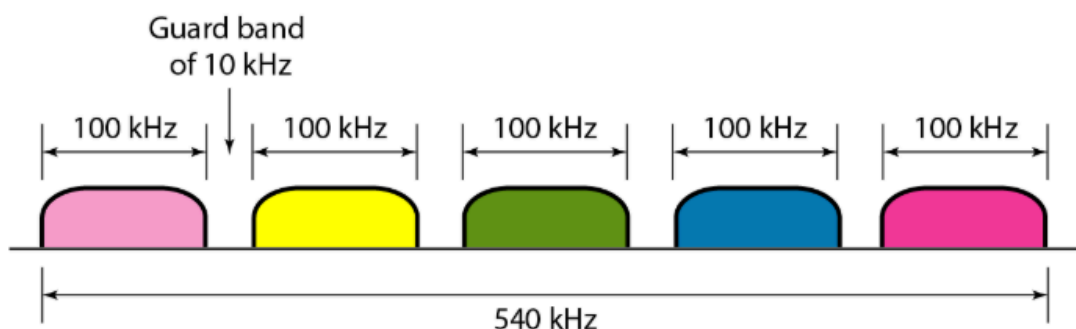


Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

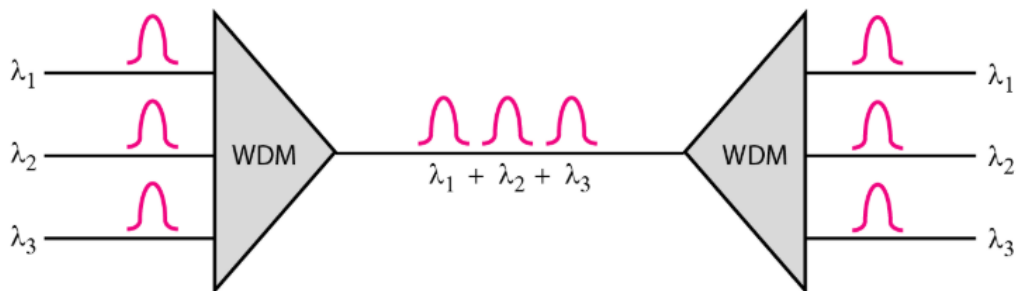
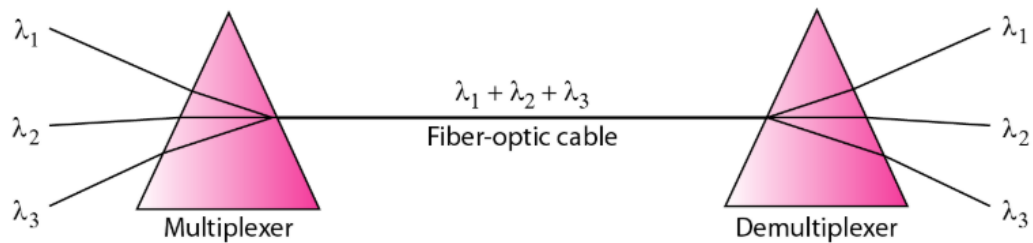
For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$



1.3 WDM

WDM is an analog multiplexing technique to combine optical signals.



Q. Describe Analog And Digital Hierarchy .

Figure 6.9 *Analog hierarchy*

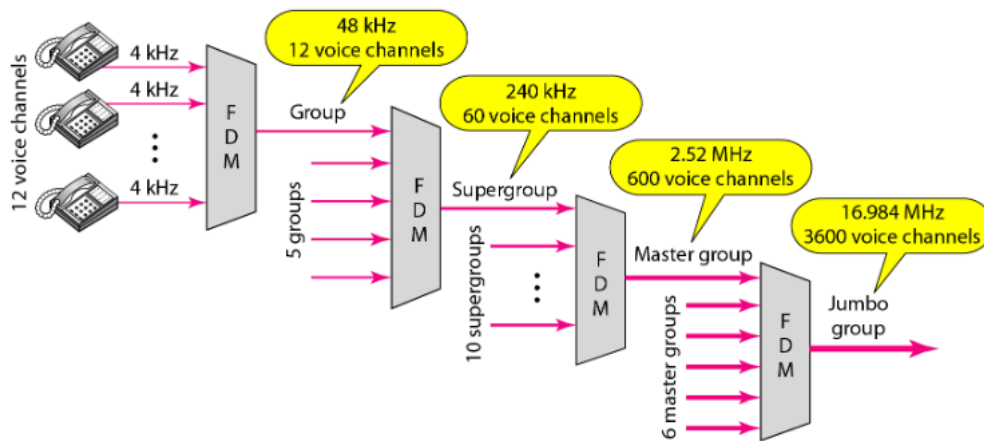
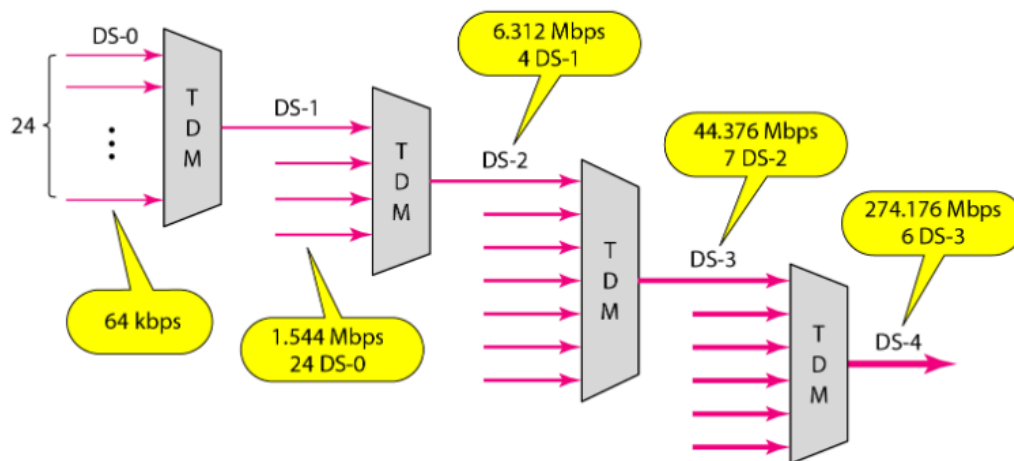


Figure 6.23 *Digital hierarchy*



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6-1 SPREAD SPECTRUM

In spread spectrum (SS), we combine signals from different sources to fit into a larger bandwidth, but our goals are to prevent eavesdropping and jamming. To achieve these goals, spread spectrum techniques add redundancy.

Topics discussed in this section:

Frequency Hopping Spread Spectrum (FHSS)

Direct Sequence Spread Spectrum Synchronous (DSSS)

Figure 6.28 Frequency hopping spread spectrum (FHSS)

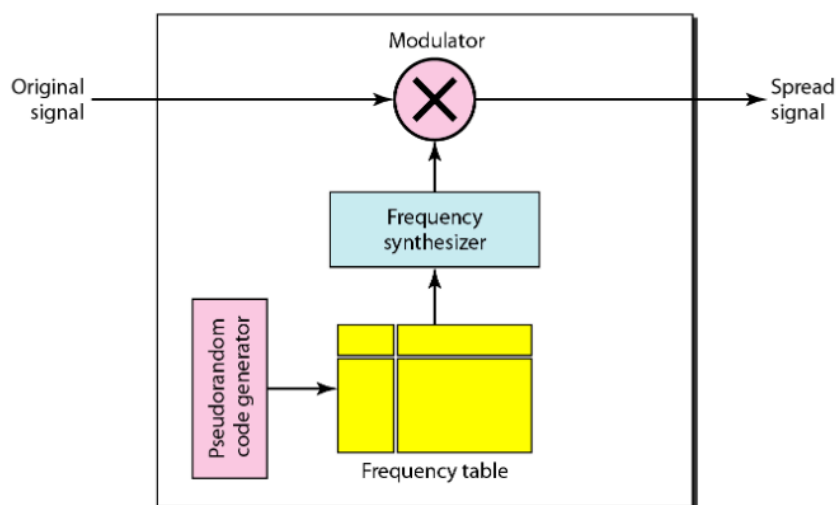


Figure 6.32 *DSSS*

